

FAQ: G2 AI Builder Assignment — The Lantern Desk

General Understanding

Q1. What exactly is this assignment asking me to build? You are building a command-line AI assistant called Lantern that helps a fictional publishing team (Lantern Press) work with files in a project folder. It should be able to search documents, answer questions from them, generate new documents, and stay safely within boundaries — all through natural-language conversation.

Q2. Do I need to build a UI or a web app? No. A graphical interface is explicitly not required. Lantern must be a CLI (command-line interface) tool only.

Q3. What programming language should I use? The assignment does not specify one. You may use any language that can call the Anthropic API. Python is the most common choice given the available SDKs, but other languages are acceptable.

Q4. Is this a chatbot or an agent? What's the difference here? It's an agent. Unlike a basic chatbot that just generates text, Lantern must be able to take actions — reading files, creating files, organising the folder — based on the request, and chain multiple actions together as part of a single task.

Using the Anthropic API

Q5. Is the API key provided in the document safe to use? The API key in the document is provided for assignment purposes. Do not commit it to a public repository (e.g., GitHub). Load it via an environment variable (e.g., `ANTHROPIC_API_KEY`) and exclude it from your submission ZIP.

The Confidence Level System

Q6. What are the four confidence levels and where do they apply? The levels — NONE, LOW, MEDIUM, HIGH — represent how strongly human approval is needed before Lantern acts:

- **NONE:** Clear, safe request. Act immediately.
- **LOW:** Minor ambiguity. Ask one short clarifying question or state a reasonable assumption.
- **MEDIUM:** Multiple valid interpretations. Show 2–3 options and let the user choose.

- **HIGH:** Destructive, restricted, or policy-conflicting action. Explain the consequence and require an explicit decision.

Q7. Who decides the confidence level — the model or my code? The model should reason about the confidence level as part of its response logic. You can prompt it to always output the confidence level before acting, and your code surfaces it to the user accordingly.

Sandbox & Safety

Q8. What exactly counts as "outside the working directory"? Anything that resolves to a path above the directory where Lantern was started. This includes `../`, absolute paths like `/etc/passwd`, symlinks pointing outside, and paths the user describes as "temporary" but that sit elsewhere on the machine.

Policy Documents

Q9. How does Lantern know which files are policy files vs. regular documents? You decide the naming or location convention and document it. Common approaches: a dedicated `/policies/` folder, or filenames beginning with `POLICY_`. Make sure Lantern checks relevant policies before any affected action.

Q10. Do policy files count toward the 10 required source documents? No. The assignment explicitly says policy files do not count toward the minimum of ten source documents.

Workspace Setup

Q11. What is the minimum workspace I need to create?

- At least 5 folders (research notes, meeting records, drafts, reference material, etc.)
- At least 2 documents per folder
- At least 10 source documents total
- Each document must have at least 300 words of meaningful text
- At least 1 policy document (doesn't count toward the 10)
- At least 1 output folder

Q12. Can I generate the document content with AI? Yes. The assignment says "you may create the text synthetically." Just make sure it's meaningful (not random filler) and coherent with a publishing/editorial theme.

Q13. What file formats are supported? `.txt`, `.md`, `.docx`, and `.pdf`. Any combination is fine. OCR and image handling are explicitly out of scope.

Q14. Does the output folder count toward the five required source folders? No. Output folders are separate and do not count toward the five source folders.

Submission

Q15. What goes in the submission ZIP?

- Application source code
- The example workspace (all files needed to reproduce test runs)
- Automated tests
- Recorded test run transcripts
- README.md and SETUP.md

Q16. What goes in README.md vs. SETUP.md?

- **README.md:** What Lantern does and how its design works — focus on core functionality and the safety/security strategy.
- **SETUP.md:** Step-by-step instructions to install dependencies, configure the environment, run Lantern in the example workspace, and run the automated tests.